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DATA DESCRIPTOR

An Integrated Dataset of Metastatic Breast Cancer to the Brain with Imaging, Radiomics, and Tumor Genetics

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This study introduces a unique magnetic resonance imaging dataset focusing on metastatic breast cancer to the brain, a significant clinical challenge in cancer treatment. Comprising 297 T1-weighted post-contrast images from 165 patients, this dataset from the University of Minnesota Medical Center is the first dedicated to breast cancer brain metastases. This collection includes expert-reviewed lesion segmentations with original image files, genetic markers, and an extensive array of tumor-derived radiomic features. The dataset's uniqueness lies in its detailed focus on metastatic breast cancer to the brain—offering a rich resource for advanced image-based tumor phenotyping and the vast potential for radiogenomic-based predictions based on machine learning model development. The inclusion of clinician-reviewed tumor segmentations and radiomic features, encompassing shape and texture characteristics, enhances the dataset's utility. This dataset aims to facilitate a deeper understanding of breast cancer metastasis to the brain, promote advancements in precision medicine, and improve patient care.

Background & Summary

Brain metastases are a significant clinical challenge in the treatment of cancer, with 15–25%¹ being due to breast cancer². Up to 40% of all patients suffering from metastatic tumors develop brain metastases³. Moreover, approximately 14% of patients with metastatic breast cancer experience brain metastases⁴. Early detection and treatment of brain metastases from breast cancer directly influences outcome^{5–7}. Thus, methods to accurately and rapidly diagnose based on radiographic features are an area of great interest to optimize the time to diagnosis and treatment.

Breast cancer metastases to the brain present a highly heterogeneous disease pathology⁸—as evidenced by their variable radiographic phenotypes in magnetic resonance imaging (MRI) scans⁹. This heterogeneity manifests as unique intensity profiles across the tumor mass, reflecting differences in underlying tumor biology, microenvironment and eventual therapeutic response¹⁰. Recent advancements in clinical practice, including the expanded use of stereotactic radiosurgery (SRS), show promise in improving patient outcomes¹¹. SRS is increasingly being applied not only as a standalone treatment but also as an adjunct to surgical resection^{12,13}—signifying an important development in the management of metastatic disease. Simultaneously, the field of radiomics, fueled by breakthroughs in machine learning tools, is evolving rapidly^{9,14–16}. Radiomics augments traditional clinical diagnostics by extracting an array of features from radiographic images, leading to advanced image-based tumor phenotyping and offering an innovative perspective on disease understanding.

Despite the growing interest in the subject of radiomics, most available datasets do not focus on metastatic lesions. Compared to datasets that encompass a broader spectrum of brain malignancies^{17–19}, ours offers a targeted and in-depth exploration of this particular site-specific subset, allowing for more

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Collection Name	# of Subjects	Pathology	Data types	Supporting data
Brain-TR-GammaKnife	47	Lung, Breast, Melanoma, Renal, Ovarian, Urothelial, Esophageal, Uterine	Other, MR, RTSTRUCT, RTDOSE, Segmentation, Demographic, Diagnosis	Clinical, Image Analyses, Software/Source Code
Pretreat-MetsToBrain-Masks	200	Breast Cancer, Small Cell Lung Cancer, Non-small Cell Lung Cancer, Melanoma	MR, Segmentation, Demographic, Diagnosis, Measurement, Follow-Up	Clinical, Image Analyses, Software/Source Code
GammaKnife-Hippocampal	198 (metastasis specifically)	Vestibular schwannoma, trigeminal neuralgia, metastatic disease	MR, RTDOSE, RTPLAN, RTSTRUCT, CT, REG	Images only
BCBM-RadioGenomics	165	Breast Cancer	MR, Segmentation, Radiomic Feature, Demographic, Molecular Test, Other	Radiomic features, Clinical, Analyses Software/Source Code

Table 1. A comparison of brain metastases-focused collections in The Cancer Imaging Archive (TCIA).

Age (Average $\pm$ STD)	58.3 $\pm$ 10.1
Gender (Females/Males)	163/0
ER status (+/-/NA)	66/71/26
PR status (+/-/NA)	46/89/28
HER2 status (+/-/NA)	70/65/28
Average # of sessions per patient	1.6

Table 2. Clinical demographics including age, gender, molecular status (ER/PR/HER2), and average number of sessions per patient.

nuanced analyses. To our knowledge, this dataset is the first dedicated specifically to metastatic breast cancer to the brain and is the largest, organized collection of metastatic lesions to the brain. Furthermore, the availability of a multitude of tumor-derived radiomic features in our dataset, meticulously computed from clinician-reviewed segmentations, ensures a richness in the data that is not found in other collections. Merging radiomic features with genetic data makes this dataset ripe for supervised learning tasks such as non-invasive genotyping or unsupervised methods such as principal component analysis at the lesion-level. As we provide image segmentations at the level of each lesion, we encourage re-analyses and even derivations of new radiomic features. Accurate tumor segmentation is a critical step in generating such high-quality radiomic data. Our dataset is unique in its inclusion of clinician-derived tumor segmentations by board-certified neurosurgeons and radiation oncologists. These detailed segmentations serve as the basis for the radiomic features, which include shape and texture characteristics²⁰. In alignment with the FAIR (Findable, Accessible, Interoperable, Re-usable) principle, our dataset is not only open-source but also designed for easy access, interoperability, and reusability²¹. Table 1 lists attributes of collections in The Cancer Imaging Archive (TCIA) which include brain metastases.

Our hope is that this dataset will facilitate deeper biological understanding of breast cancer and its metastasis to the brain, promote advancements in image-based tumor phenotyping and genotyping, and ultimately contribute to improving patient care and the realization of precision medicine in this disease domain.

Methods

Patient selection and imaging data acquisition. This study utilized a total of 297 3D T1-weighted post-contrast images from 165 unique patients with metastatic breast cancer. Histologic features were obtained in one of several ways: lymph node biopsy, primary breast sampling, or via biopsy of a metastatic site (either intracranial or elsewhere). All patients were treated at the University of Minnesota Medical Center (UMMC). The images were acquired as high resolution, 3D T1 post-contrast MRI scans that were part of routine clinical care. The dataset presented here contains a mixture of segmentations encompassing the tumor core, postoperative tumor bed, and key neighboring at-risk structures. Table 2 gives a summary of clinical demographics and molecular status. Table 3 gives the same data at the lesion level. All patient identifiers were removed from the dataset to uphold patient confidentiality. Each patient is assigned a unique identifier (BCBM-RadioGenomics-X), with subsequent follow-up imaging denoted as BCBM-RadioGenomics-X-#. For example, for patient BCBM-RadioGenomics-161, initial scan is denoted as folder BCBM-RadioGenomics-161-0' and first follow-up denoted as folder 'BCBM-RadioGenomics-161-1. Figure 1 shows a directory tree showing folder organization and contents.

This study was approved as a retrospective analysis by the University of Minnesota Institutional Review Board (STUDY00007985). Given the retrospective nature of the study and the use of de-identified patient data, a requirement for true informed consent was waived by the IRB. This waiver was granted on the basis that the research poses minimal risk to the individuals whose data were analyzed, and the data were used exclusively for research purposes, ensuring confidentiality and compliance with applicable privacy laws. All methods were carried out in accordance with relevant guidelines and regulations.

Image preprocessing. Prior to any analysis, the images underwent preprocessing steps to ensure their quality and comparability. The first stage was brain extraction using HD-BET—a pre-trained neural network developed as part

Features	Run parameters		
original_shape_Elongation	Category	Parameter	Default Value
original_shape_Flatness	General	resampledPixelSpacing	None
original_shape_LeastAxisLength	General	interpolator	sitkBSpline
original_shape_MajorAxisLength	General	normalize	FALSE
original_shape_Maximum2DDiameterColumn	General	normalizeScale	1
original_shape_Maximum2DDiameterRow	General	removeOutliers	None
original_shape_Maximum2DDiameterSlice	General	preCrop	FALSE
original_shape_Maximum3DDiameter	General	padDistance	5
original_shape_MeshVolume	Mask/ROI	label	1
original_shape_MinorAxisLength	Mask/ROI	geometryTolerance	1.00E-06
original_shape_Sphericity	Image Types	Original	TRUE
original_shape_SurfaceArea	Image Types	Wavelet	FALSE
original_shape_SurfaceVolumeRatio	Image Types	LoG	FALSE
original_shape_VoxelVolume	Image Types	Square	FALSE
original_firstorder_10Percentile	Image Types	SquareRoot	FALSE
original_firstorder_90Percentile	Image Types	Logarithm	FALSE
original_firstorder_Energy	Image Types	Exponential	FALSE
original_firstorder_Entropy	Image Types	Gradient	FALSE
original_firstorder_InterquartileRange	Image Types	LBP2D	FALSE
original_firstorder_Kurtosis	Image Types	LBP3D	FALSE
original_firstorder_Maximum	Feature Classes	firstorder	TRUE
original_firstorder_MeanAbsoluteDeviation	Feature Classes	shape	TRUE
original_firstorder_Mean	Feature Classes	glcm	TRUE
original_firstorder_Median	Feature Classes	glrlm	TRUE
original_firstorder_Minimum	Feature Classes	glszm	TRUE
original_firstorder_Range	Feature Classes	gldm	TRUE
original_firstorder_RobustMeanAbsoluteDeviation	Feature Classes	ngtdm	TRUE
original_firstorder_RootMeanSquared	Texture Parameters	binWidth	25
original_firstorder_Skewness	Texture Parameters	force2D	FALSE
original_firstorder_TotalEnergy	Texture Parameters	force2Ddimension	0
original_firstorder_Uniformity			
original_firstorder_Variance			
original_glcm_Autocorrelation			
original_glcm_ClusterProminence			
original_glcm_ClusterShade			
original_glcm_ClusterTendency			
original_glcm_Contrast			
original_glcm_Correlation			
original_glcm_DifferenceAverage			
original_glcm_DifferenceEntropy			
original_glcm_DifferenceVariance			
original_glcm_Id			
original_glcm_Idm			
original_glcm_Idmn			
original_glcm_Idn			
original_glcm_Imc1			
original_glcm_Imc2			
original_glcm_InverseVariance			
original_glcm_JointAverage			
original_glcm_JointEnergy			
original_glcm_JointEntropy			
original_glcm_MCC			
original_glcm_MaximumProbability			
original_glcm_SumAverage			
original_glcm_SumEntropy			
original_glcm_SumSquares			
original_gldm_DependenceEntropy			
Continued			

Features	Run parameters		
original_gldm_DependenceNonUniformity			
original_gldm_DependenceNonUniformityNormalized			
original_gldm_DependenceVariance			
original_gldm_GrayLevelNonUniformity			
original_gldm_GrayLevelVariance			
original_gldm_HighGrayLevelEmphasis			
original_gldm_LargeDependenceEmphasis			
original_gldm_LargeDependenceHighGrayLevelEmphasis			
original_gldm_LargeDependenceLowGrayLevelEmphasis			
original_gldm_LowGrayLevelEmphasis			
original_gldm_SmallDependenceEmphasis			
original_gldm_SmallDependenceHighGrayLevelEmphasis			
original_gldm_SmallDependenceLowGrayLevelEmphasis			
original_glrlm_GrayLevelNonUniformity			
original_glrlm_GrayLevelNonUniformityNormalized			
original_glrlm_GrayLevelVariance			
original_glrlm_HighGrayLevelRunEmphasis			
original_glrlm_LongRunEmphasis			
original_glrlm_LongRunHighGrayLevelEmphasis			
original_glrlm_LongRunLowGrayLevelEmphasis			
original_glrlm_LowGrayLevelRunEmphasis			
original_glrlm_RunEntropy			
original_glrlm_RunLengthNonUniformity			
original_glrlm_RunLengthNonUniformityNormalized			
original_glrlm_RunPercentage			
original_glrlm_RunVariance			
original_glrlm_ShortRunEmphasis			
original_glrlm_ShortRunHighGrayLevelEmphasis			
original_glrlm_ShortRunLowGrayLevelEmphasis			
original_glszm_GrayLevelNonUniformity			
original_glszm_GrayLevelNonUniformityNormalized			
original_glszm_GrayLevelVariance			
original_glszm_HighGrayLevelZoneEmphasis			
original_glszm_LargeAreaEmphasis			
original_glszm_LargeAreaHighGrayLevelEmphasis			
original_glszm_LargeAreaLowGrayLevelEmphasis			
original_glszm_LowGrayLevelZoneEmphasis			
original_glszm_SizeZoneNonUniformity			
original_glszm_SizeZoneNonUniformityNormalized			
original_glszm_SmallAreaEmphasis			
original_glszm_SmallAreaHighGrayLevelEmphasis			
original_glszm_SmallAreaLowGrayLevelEmphasis			
original_glszm_ZoneEntropy			
original_glszm_ZonePercentage			
original_glszm_ZoneVariance			
original_ngtdm_Busyness			
original_ngtdm_Coarseness			
original_ngtdm_Complexity			
original_ngtdm_Contrast			
original_ngtdm_Strength			

Table 3. 107 original image features from the PyRadiomics package. In the second column are the default run parameters that were used when running feature extraction.

of a large, multi-institutional collaboration among European institutions and validated on several external datasets²². Subsequently, N4 bias correction was applied to all skull-stripped images²³. This correction is a critical preprocessing step in neuroimaging analysis that corrects the intensity non-uniformity artifact prevalent in MRI data, thereby improving the quality of the images and ensuring the reliability of subsequent analyses. This task was accomplished

File Structure

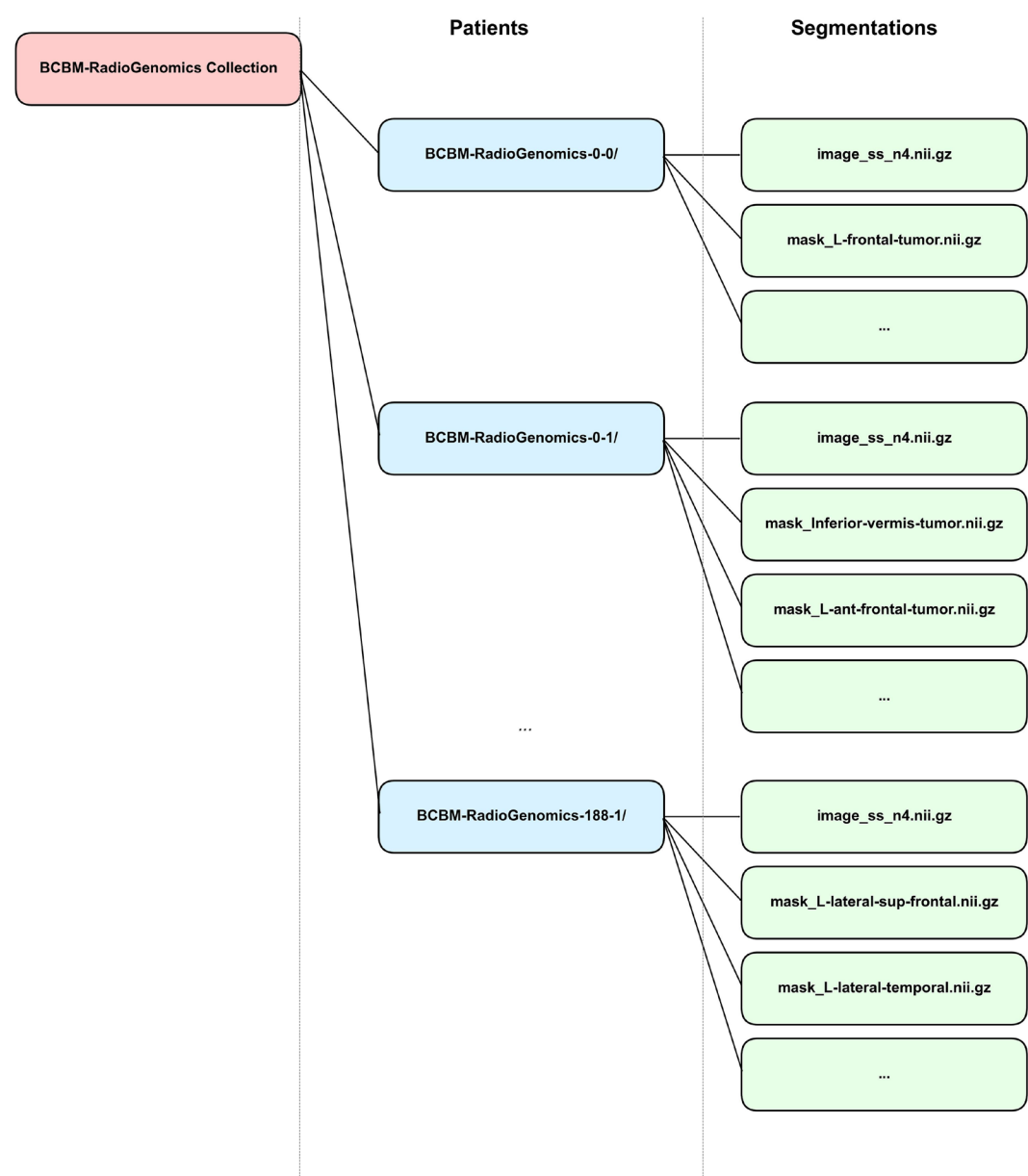


Fig. 1 The file structure of the collection with sample patients included. Brain extracted and N4-normalized images are included in each patient folder along with respective segmentations which map directly onto the patient image.

	HER2 status	ER status	PR status
Number of patients with data	145	147	145
Lesions	1125	1142	1135
Age (average, min-max)	57.3 (27–88)		
Gender (Female)	100%	100%	100%
Positive	850	825	266

Table 4. Patient-level and lesion-level data with demographics and molecular status.

using the Python module ANTsPy (Advanced Normalization Tools), a wrapper for the C++ ANTs library²⁴. Figure 1 shows our workflow in a step-by-step fashion.

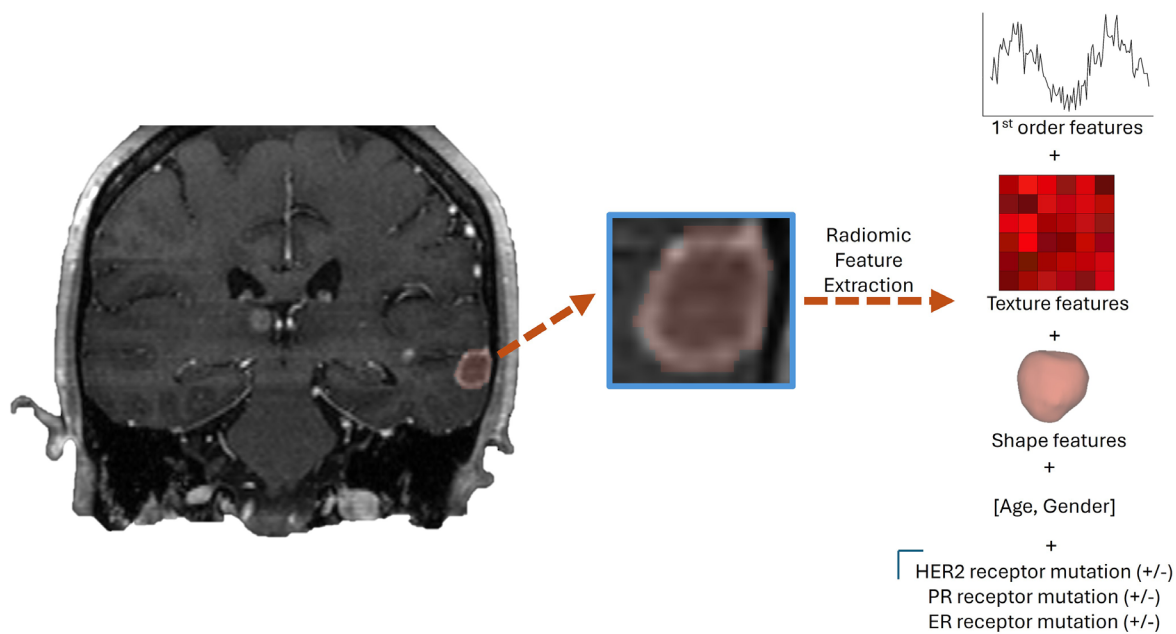


Fig. 2 A sample patient with a dural based metastasis used as a graphical representation of the workflow from original image to brain extraction, N4 inhomogeneity correction, segmentation and finally the extraction of radiomic features. Radiomic information is then combined with clinical demographic data (age, gender) and molecular status information (ER/PR/HER2).

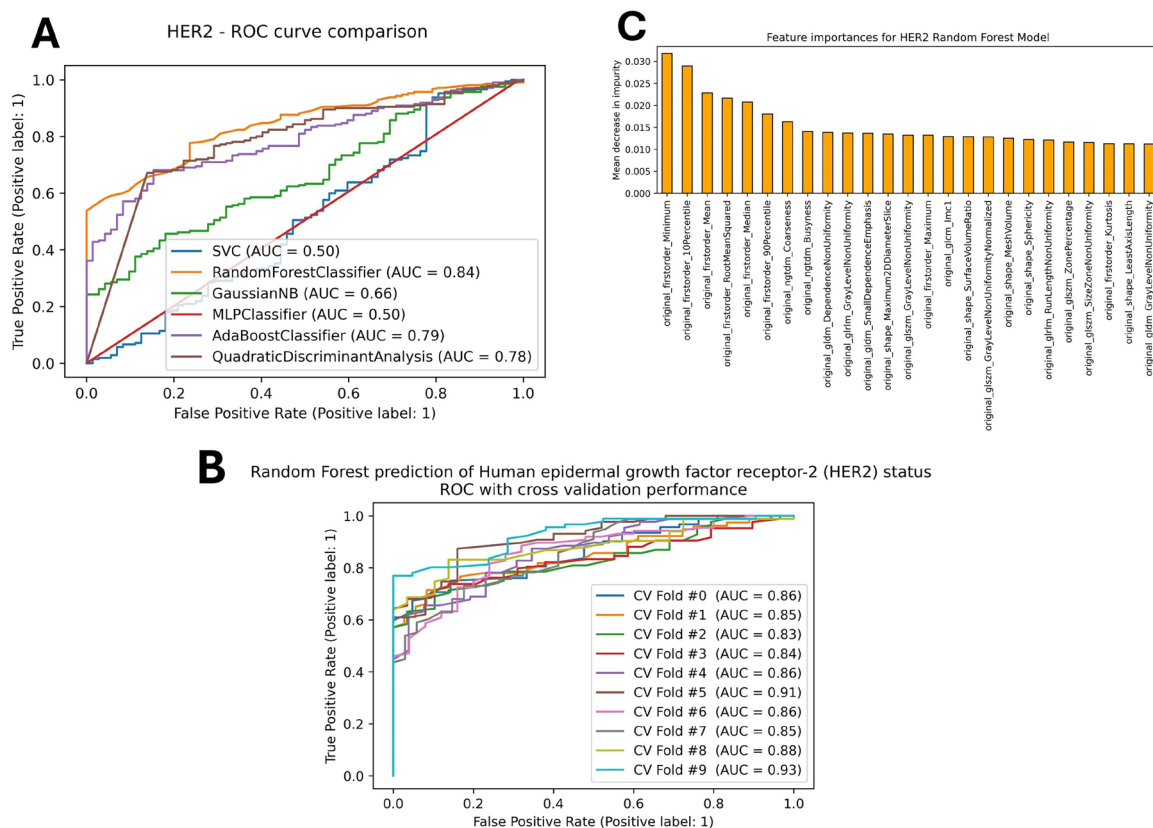


Fig. 3 (A) A comparison of multiple machine learning model ROC curves for HER2 mutation prediction demonstrating the superiority of the random forest classifier. (B) ROC curve performance for each cross validation iteration (listed as CV Fold #). (C) An ordered list of relative feature importances for all radiomic features.

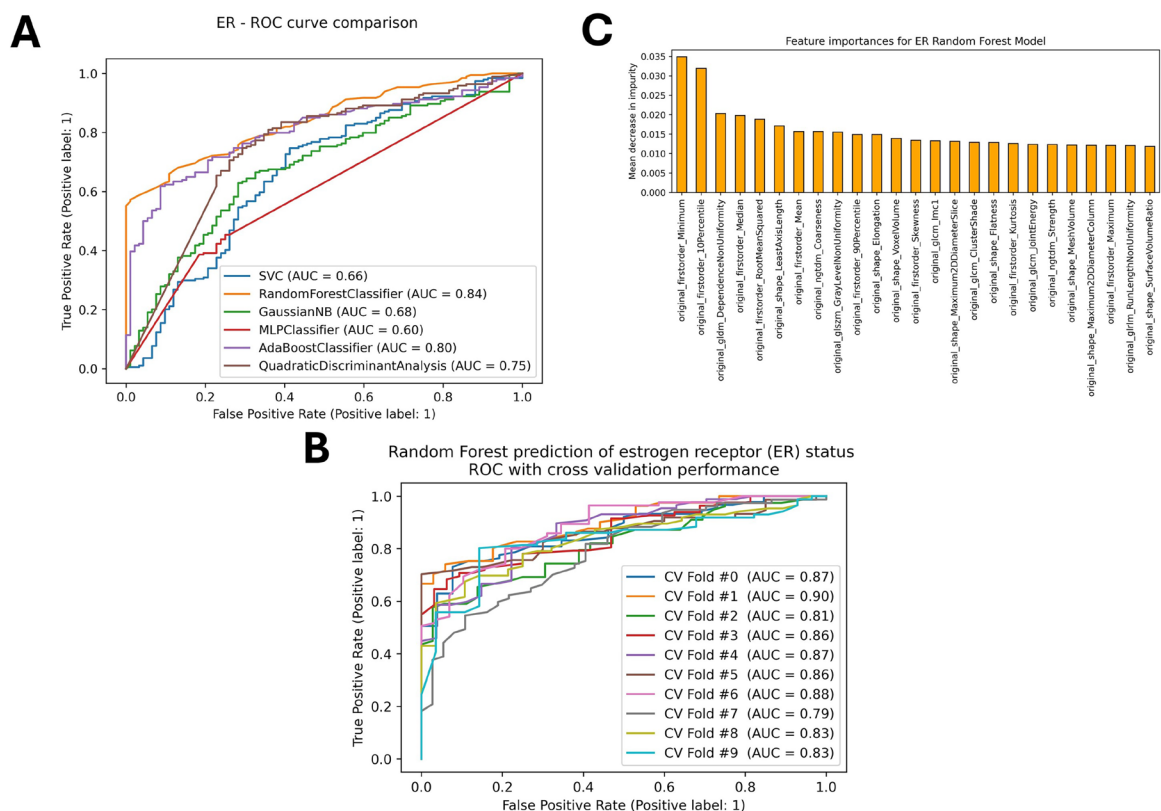


Fig. 4 (A) A comparison of multiple machine learning model ROC curves for ER mutation prediction demonstrating the superiority of the random forest classifier. (B) ROC curve performance for each cross validation iteration (listed as CV Fold #). (C) An ordered list of relative feature importances for all radiomic features.

Tumor segmentation. The segmented images included in this dataset were produced via a semi-automated, consensus segmentation method. A board-certified neurosurgeon and a radiation oncologist utilized segmentation software within the SRS system for slice-by-slice delineation of the metastatic brain tumors and neighboring structures at risk.

Radiomic feature extraction. Radiomic features were derived from the segmented regions using the PyRadiomics library, a flexible Python package widely used for the extraction of radiomic features from medical imaging data²⁵. This library was used to extract a wide array of features that capture the shape and texture characteristics of structures of interest and their filtered analogs (Wavelet and Laplacian of Gaussian). Shape features describe the two and three-dimensional form of the segmented areas. Texture features capture the patterns and distribution of pixel intensities within the region, indicative of intratumoral heterogeneity²⁶. These radiomic features provide a comprehensive representation of the tumors, facilitating in-depth analysis and machine learning model development. A complete list of radiomic features and default run parameters are listed in Table 4.

Data Records

All data records are stored in the TCIA²⁷. Pre-processed images and segmentations are stored in the Neuroimaging Informatics Technology Initiative (NIfTI) format. Clinical, molecular, scan parameters, and radiomic features are also freely available in XSLX format. A complete list of scan parameters for each scan and a table of radiomic features for all lesions can be found as Supplementary File 1 and Supplementary File 2, respectively.

Technical Validation

Segmentations were produced through a semi-automated, consensus approach involving a board-certified neurosurgeon and a radiation oncologist during SRS planning. These segmentations enable direct analysis of tumor regions, radiotherapy margins, and surrounding structures in order to extract radiomic information.

With the available data, we developed a radiomics-based classifier for the prediction of Estrogen Receptor (ER), Progesterone Receptor (PR), Human Epidermal Growth Receptor (HER2) status using an 80:20 training-test split. In model training and testing, lesions were treated as independent samples and their labels attributed to the patient's ER/PR/HER2 status. A binary classifier was trained on each dataset with available genetic status. Multiple models were trained including: support vector machine classifier (SVC), Gaussian

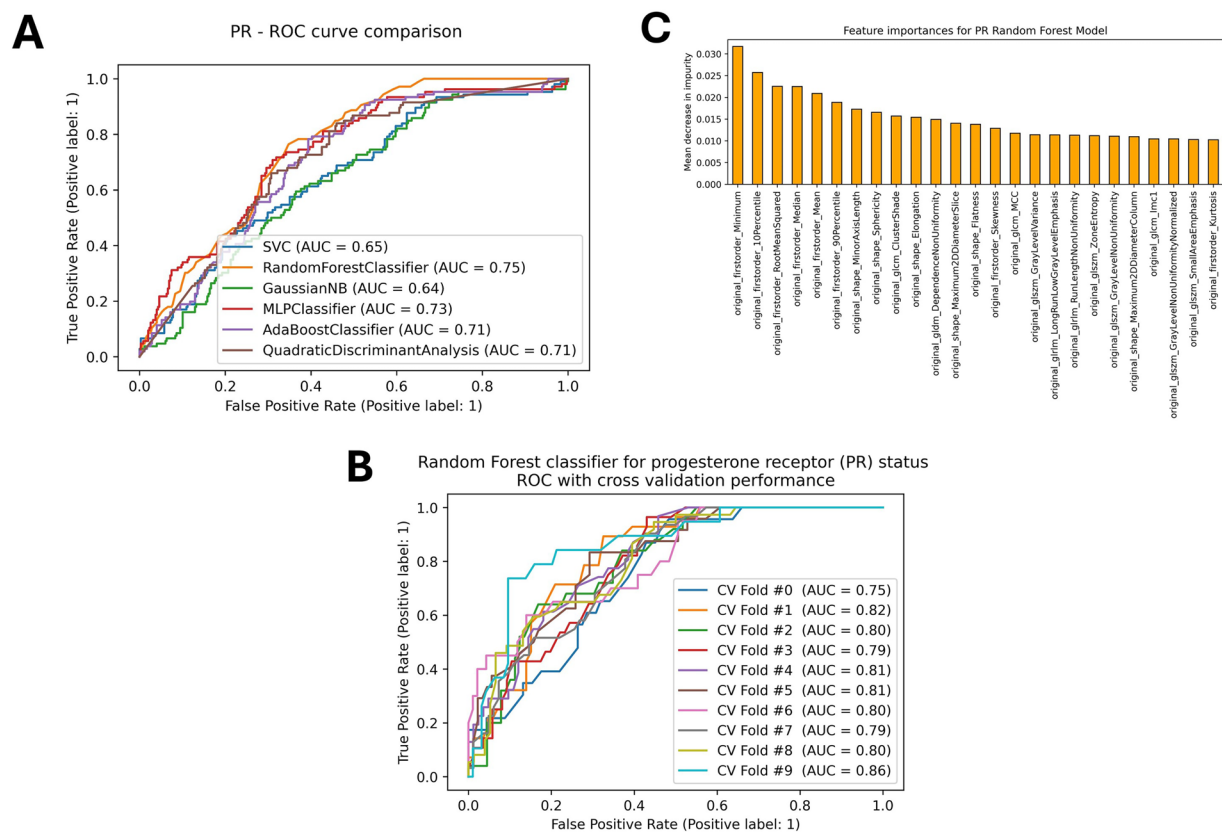


Fig. 5 (A) A comparison of multiple machine learning model ROC curves for ER mutation prediction demonstrating the superiority of the random forest classifier. (B) ROC curve performance for each cross validation iteration (listed as CV Fold #). (C) An ordered list of relative feature importances for all radiomic features.

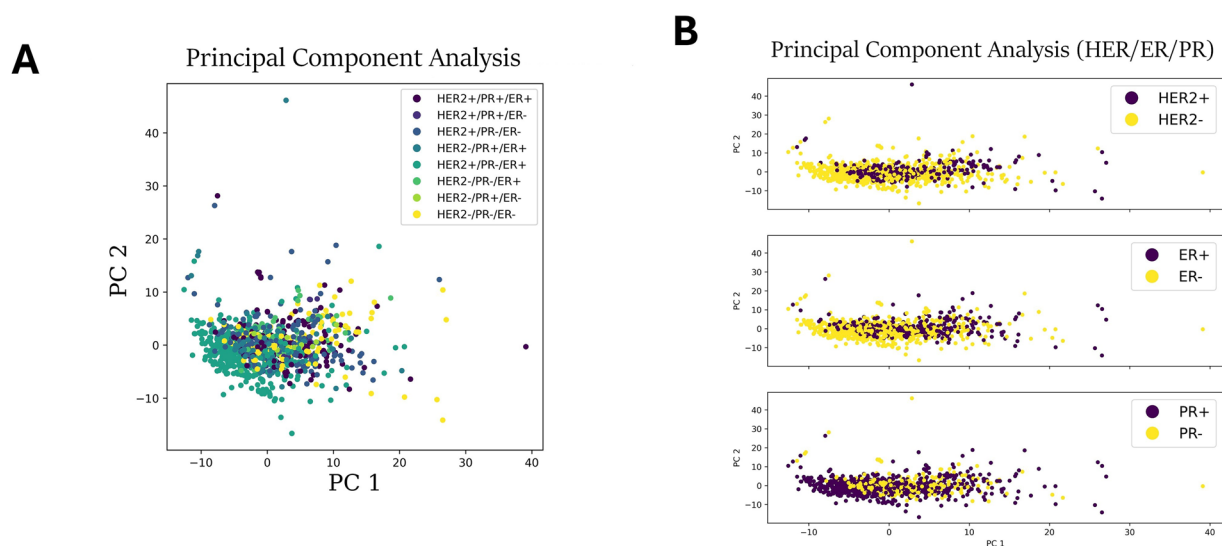


Fig. 6 (A) Principal component analysis (PCA) demonstrating the first two components for all lesions with ER/PR/HER2 status known using radiomic features. (B) Individual PCA plots for ER/PR/HER2 using radiomic features.

Naive-Bayes (GNB), multi-layer perceptron (MLP), AdaBoost classifier, quadratic discriminant analysis and random forest (RF). Evaluation was done using five-fold cross-validation. Each model was evaluated based on accuracy, precision, recall, F1 score, and the area under the receiver operating curve (AUC). Receiver operating curves (ROC) were generated for the most successful model.

In HER2 mutation prediction, a random forest classifier showed the best overall performance (F1: 0.87, Recall: 93.2%, Accuracy: 77.6%, AUC: 0.77). Figure 3A–C demonstrates the robust performance of the random forest classifier and the relative feature importances. In the prediction of ER mutation, similarly, a random forest classifier performed the best (F1: 0.84, Accuracy: 75.9%, Precision: 82%, Recall: 86.2%, AUC: 0.81). Figure 4A–C shows complete model performance and ordered bar plot of relative feature importances. For PR mutation prediction, a random forest classifier again showed the best performance overall (F1: 0.40, Accuracy: 78.5%, Precision: 60.3%, Recall: 30.0%, AUC: 0.75). Figure 5A–C shows model performance comparison and plot of relative radiomic feature importances.

We also explored an unsupervised approach to investigate the relationship between radiomic features and genetic status. We performed principal component analysis (PCA) on the full radiomic features at the lesion level, and colored each lesion based on its ER/PR/HER2 status, displayed in Fig. 6A. We also performed a separate PCA using ER positive vs negative samples, PR positive vs negative samples, and HER2 positive vs negative samples individually shown in Fig. 6B.

Usage Notes

The complete imaging data is organized into folders for each subject which contains the original image, segmentation files. A separate XLSX file contains radiomic features data for each lesion and subject. A third file contains the complete clinical demographics and genetic status of ER/PR/HER2. Image segmentations, radiomic data, and clinical demographic data are hosted at TCIA and are freely available for download under a CC BY 4.0 License. Images and segmentations can be viewed using any available software that can view NIfTI files (Slicer²⁸, ITK-SNAP²⁹).

Data availability

Our complete collection is made available at <https://www.cancerimagingarchive.net/collection/BCBM-RadioGenomics/>.

Code availability

The code used for image preprocessing, segmentation, and radiomic feature extraction in this article can be found at <https://github.com/birra-taha/BreastCancer>.

Received: 9 February 2025; Accepted: 30 September 2025;

Published online: 20 November 2025

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Acknowledgements

The authors acknowledge the support of the University of Minnesota Medical Center for data access and MRI acquisition. The authors thank the Gamma Knife Radiosurgery team for their assistance in tumor segmentation.

Author contributions

B.R.T. conceptualized the study, contributed to conceptualization, data collection, segmentation verification, literature review, and manuscript writing. D.W. was responsible for data acquisition, and manuscript editing. L.T.S. performed data acquisition, literature review, manuscript writing and editing. M.K. conducted data acquisition and manuscript editing. A.S.V. contributed to methodology development, and provided critical manuscript review. Y.W. supervised the project, provided access to data, manuscript review. J.Y., M.H., D.P.D., B.A.S., L.S., M.C.P. was responsible for data acquisition and provided critical manuscript review. All authors contributed to data interpretation and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41597-025-06066-6>.

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